Understanding World Models in Machine Learning and AGI

Pipi Hu

Beijing Institute of Mathematical Sciences and Applications

October 7, 2025

Feynman's words

What I cannot create, I do not understand.

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

The Concept of World Models

A **world model** is an internal representation that an AI system constructs to simulate and understand its environment.

- Abstraction: Extract high-level features from raw sensory data
- Prediction: Forecast future states and rewards
- Simulation: Internally simulate virtual scenarios
- Interaction: Infer intentions and beliefs of other agents

Key Insight: World models enable AI systems to reason about the world without direct interaction, crucial for achieving AGI.

Historical Context: From Reinforcement Learning to AGI

1. **Classical RL**: Direct interaction with environment

$$\pi^*(s) = \arg\max_a \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid s_0 = s, a_0 = a\right] \tag{1}$$

2. **Model-based RL**: Learn environment dynamics

$$\hat{p}(s_{t+1}, r_{t+1} \mid s_t, a_t) \approx p(s_{t+1}, r_{t+1} \mid s_t, a_t) \tag{2}$$

3. **World Models**: Internal simulation and planning

$$\text{World Model} = \{\text{Encoder}, \text{Dynamics}, \text{Decoder}\} \tag{3}$$

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Formal Definition of World Models

A world model $\mathcal{W}$ consists of three components:

1. **Encoder**: $z_t = \text{Enc}(s_t)$ - maps observations to latent states
2. **Dynamics**: $z_{t+1} = f(z_t, a_t)$ - predicts next latent state
3. **Decoder**: $\hat{s}_t = \text{Dec}(z_t)$ - reconstructs observations

Objective: Learn a compressed representation that captures the essential dynamics of the environment.

The World Model Learning Problem

Given a sequence of observations and actions $\{(s_t, a_t)\}_{t=1}^{T}$, we want to learn:

$$\min_{\theta} \mathcal{L}_{\text{world}} = \mathbb{E}\left[\sum_{t=1}^{T} \|\hat{s}_t - s_t\|^2 + \lambda \mathcal{L}_{\text{consistency}}\right] \tag{4}$$

where $\mathcal{L}_{\text{consistency}}$ ensures the learned dynamics are consistent with the true environment dynamics.

Key Challenge: The dynamics model must be both accurate and generalizable to unseen scenarios.

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Variational World Models

Following the variational autoencoder framework, we can formulate world models as:

$$p(z_{1:T} \mid s_{1:T}, a_{1:T}) = \prod_{t=1}^{T} p(z_t \mid z_{t-1}, a_{t-1}) \tag{5}$$

Variational Lower Bound:

$$\log p(s_{1:T} \mid a_{1:T}) \geq \mathbb{E}_{q(z_{1:T})} \left[\sum_{t=1}^{T} \log p(s_t \mid z_t) - \mathrm{KL}(q(z_t \mid z_{t-1}, a_{t-1}) \| p(z_t \mid z_{t-1}, a_{t-1})) \right] \tag{6}$$

This formulation allows for uncertainty quantification and better generalization.

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Outline

BIMSA
北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Model-Based Reinforcement Learning with World Models

Traditional model-based RL learns environment dynamics:

$$\hat{p}(s_{t+1}, r_{t+1} \mid s_t, a_t) \approx p(s_{t+1}, r_{t+1} \mid s_t, a_t) \tag{7}$$

World Model Approach: Learn dynamics in latent space

$$\hat{z}_{t+1} = f_\theta(z_t, a_t), \quad \hat{r}_{t+1} = g_\theta(z_t, a_t) \tag{8}$$

Advantages:

- More efficient learning in high-dimensional spaces
- Better generalization to unseen states
- Enables planning in latent space

Planning with World Models

Once we have a world model, we can plan by:

1. **Rollout Planning**: Simulate trajectories in latent space

$$z_{t+1} = f(z_t, a_t), \quad \hat{r}_{t+1} = g(z_t, a_t) \tag{9}$$

2. **Value Function Learning**: Learn $V(z)$ in latent space

$$V(z_t) = \mathbb{E}\left[\sum_{k=0}^{H} \gamma^k r_{t+k} \mid z_t\right] \tag{10}$$

3. **Policy Optimization**: Update policy based on simulated experience

Key Insight: Planning in compressed latent space is computationally more efficient than planning in raw observation space.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Dreamer Algorithm: A Case Study

The Dreamer algorithm demonstrates world model learning:

1. **Representation Learning**:
$$z_t \sim q_\phi(z_t \mid s_t, z_{t-1}, a_{t-1}) \tag{11}$$
2. **Dynamics Learning**:
$$\hat{z}_{t+1} \sim p_\theta(z_{t+1} \mid z_t, a_t) \tag{12}$$
3. **Reward Prediction**:
$$\hat{r}_{t+1} = r_\theta(z_t, a_t) \tag{13}$$
4. **Policy Learning**: Learn policy in latent space
$$\pi_\psi(a_t \mid z_t) = \text{softmax}(Q_\psi(z_t, a_t)) \tag{14}$$

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

The Role of World Models in AGI

World models are fundamental for achieving AGI because they provide:

1. **Generalization**: Transfer knowledge across domains
2. **Abstraction**: Extract high-level concepts
3. **Planning**: Reason about future scenarios
4. **Imagination**: Generate novel scenarios

AGI Requirements:

$$\text{AGI} = \{\text{World Model}, \text{Reasoning}, \text{Learning}, \text{Planning}\} \tag{15}$$

Hierarchical World Models

For complex environments, hierarchical world models are necessary:

$$\mathcal{W}_{\text{hier}} = \{\mathcal{W}_1, \mathcal{W}_2, \ldots, \mathcal{W}_L\} \tag{16}$$

where each level $\mathcal{W}_i$ operates at different temporal and spatial scales.

Multi-scale Dynamics:

$$z_t^{(i)} = f^{(i)}(z_{t-1}^{(i)}, a_{t-1}^{(i)}, z_t^{(i-1)}) \tag{17}$$

This enables reasoning at multiple levels of abstraction, crucial for AGI.

World Models and Theory of Mind

For social AGI, world models must include models of other agents:

$$\mathcal{W}_{\text{social}} = \{\mathcal{W}_{\text{self}}, \mathcal{W}_{\text{others}}, \mathcal{W}_{\text{interaction}}\} \tag{18}$$

Other Agents' Models:

$$\hat{a}_j^{(t+1)} = \pi_j(z_j^{(t)}, \mathcal{W}_{\text{others}}) \tag{19}$$

This enables:

- Predicting other agents' actions
- Cooperative behavior
- Strategic reasoning

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Outline

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Transformer-based World Models

Recent work uses transformers for world modeling:

$$z_{t+1} = \text{Transformer}(z_{1:t}, a_{1:t}) \tag{20}$$

Advantages:

- Long-range dependencies
- Parallel processing
- Attention mechanisms for relevant information

Challenges:

- Computational complexity
- Memory requirements
- Training stability

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Diffusion World Models

Inspired by diffusion models, we can formulate world models as:

$$p(z_{t+1} \mid z_t, a_t) = \int p(z_{t+1} \mid z_0, a_t) p(z_0 \mid z_t) dz_0 \tag{21}$$

where the diffusion process models the uncertainty in state transitions.

Benefits:

- Natural uncertainty quantification
- Better handling of stochastic environments
- Improved sample efficiency

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Current Challenges in World Models

1. **Model Complexity**: Balancing expressiveness and computational efficiency
2. **Generalization**: Transferring knowledge across domains
3. **Uncertainty**: Quantifying and propagating uncertainty
4. **Scalability**: Scaling to complex, high-dimensional environments

Open Questions:

- How to learn world models from limited data?
- How to ensure world models are interpretable?
- How to combine multiple world models?

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Future Research Directions

1. **Neural-Symbolic World Models**: Combining neural networks with symbolic reasoning
2. **Continual Learning**: Learning world models that adapt over time
3. **Multi-modal World Models**: Integrating different sensory modalities
4. **World Model Composition**: Combining multiple specialized world models

Long-term Vision: World models that can:

- Learn from minimal data
- Transfer knowledge across domains
- Reason about abstract concepts
- Collaborate with other agents

Outline

World Models in Robotics

- **Sim-to-Real Transfer**: Train in simulation, deploy in real world
- **Manipulation Planning**: Plan complex manipulation tasks
- **Navigation**: Navigate in unknown environments

Key Insight: World models enable robots to reason about the physical world without direct interaction.

World Models in Game AI

- **Strategy Games**: Model opponent behavior and game dynamics
- **Real-time Strategy**: Plan complex multi-agent strategies
- **Adversarial Games**: Model adversarial environments

Examples:

- AlphaGo: Model-based planning in Go
- StarCraft AI: Strategic planning in RTS games
- Chess engines: Position evaluation and planning

World Models in Scientific Discovery

- **Physics Simulation**: Model physical laws and phenomena
- **Chemical Reactions**: Predict reaction outcomes
- **Biological Systems**: Model complex biological processes

Potential Impact: World models could accelerate scientific discovery by enabling:

- Virtual experimentation
- Hypothesis generation
- Theory validation

Outline

1. Introduction: What are World Models?
2. Mathematical Foundations of World Models
3. World Models in Reinforcement Learning
4. World Models for AGI
5. Advanced World Model Architectures
6. Challenges and Future Directions
7. Applications and Case Studies
8. **Theoretical Foundations**
9. Implementation Considerations
10. Conclusion and Future Work

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Information-Theoretic Perspective

World models can be understood through information theory:

$$I(S; Z) = H(S) - H(S \mid Z) \tag{22}$$

where $I(S; Z)$ is the mutual information between observations S and latent states Z.
Optimal World Model: Maximizes mutual information while minimizing complexity:

$$\max_{\theta} I(S; Z_\theta) - \lambda \text{Complexity}(Z_\theta) \tag{23}$$

Causal World Models

For AGI, world models must capture causal relationships:

$$p(\text{effect} \mid \text{cause}) \neq p(\text{effect} \mid \text{correlation}) \tag{24}$$

Causal Discovery: Learning causal graphs from data

$$\mathcal{G} = \{(V, E) : V \text{ variables}, E \text{ causal edges}\} \tag{25}$$

Intervention: Predicting effects of interventions

$$p(Y \mid \text{do}(X = x)) \neq p(Y \mid X = x) \tag{26}$$

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

World Models and Consciousness

Some researchers argue that world models are related to consciousness:

- **Self-Model**: Model of the agent itself
- **Attention**: Focus on relevant aspects of the world
- **Memory**: Integration of past experiences
- **Prediction**: Anticipation of future events

Open Question: Can sophisticated world models lead to artificial consciousness?

Outline

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Computational Requirements

World models have significant computational requirements:

- **Training**: Large-scale optimization problems
- **Inference**: Real-time prediction and planning
- **Memory**: Storing and retrieving world state
- **Communication**: Sharing world models between agents

Optimization Strategies:

- Distributed training
- Model compression
- Efficient architectures
- Hardware acceleration

Evaluation Metrics

How do we evaluate world models?

1. **Prediction Accuracy**: How well does the model predict future states?
2. **Planning Performance**: How well can the model be used for planning?
3. **Generalization**: How well does the model transfer to new domains?
4. **Efficiency**: How computationally efficient is the model?

Benchmarks:

- Atari games
- MuJoCo environments
- Real-world robotics tasks
- Multi-agent environments

Outline

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Summary

World models are fundamental for achieving AGI:

- **Mathematical Foundation**: Rigorous theoretical framework
- **Practical Applications**: Success in various domains
- **Research Challenges**: Many open problems remain
- **Future Potential**: Key to artificial general intelligence

Key Takeaways:

1. World models enable efficient learning and planning
2. They are crucial for generalization and transfer
3. Current methods show promise but need improvement
4. Future work should focus on scalability and robustness

Open Research Questions

1. How can we learn world models from limited data?
2. How do we ensure world models are interpretable and trustworthy?
3. How can we combine multiple world models effectively?
4. How do we handle non-stationary environments?
5. How can world models enable artificial consciousness?

Call for Research: These questions represent exciting opportunities for future research in AI and AGI.

Thank You

Questions and Discussion

World models represent a crucial step toward artificial general intelligence. The mathematical foundations are solid, but many challenges remain.

Contact: hpp@bimsa.cn

References:

- Ha & Schmidhuber (2018): World Models
- Hafner et al. (2019): Dreamer
- Chen et al. (2021): Decision Transformer
- And many more...

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications